6 *Leveraging information theory with entropy-based policies*

This chapter covers

- Entropy as an information theoretic measure of uncertainty
- Information gain as a method of reducing entropy
- Bayesian optimization policies that leverage information theory for their search

We saw in Chapter 4 that by aiming to improve from the best value that we have seen so far, we can design improvement-based Bayesian optimization (BayesOpt) policies such as *Probability of Improvement* and *Expected Improvement*. In Chapter 5, we leveraged multi-armed bandit policies to obtain *Upper Confidence Bound* and *Thompson sampling*, each of which uses a unique heuristic to balance between exploration and exploitation in the search for the global optimum of the objective function.

In this chapter, we learn about another heuristic to decision making, this time using information theory, to design BayesOpt policies that we can use in our optimization pipeline. Unlike the heuristics that we have seen (seeking improvement, optimism in the face of uncertainty, random sampling), which might seem unique to optimization-related tasks, information theory in itself is a major subfield of mathematics that has application in a wide range of topics. As we will see, by appealing to information theory, or more specifically entropy, a quantity that measures uncertainty in terms of information, we can design BayesOpt policies that seek to reduce our uncertainty about a quantity of interest in a principled, mathematically elegant manner.

The idea behind entropy-based search is quite simple: we go look at places where our information about a quantity of interest will most increase. As we will see later in the chapter, this is similar to looking for a lost remote control in the living room where the TV is, as opposed to in the bathroom.

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